SHARED EQUIPMENT. PROTECTED REPAIR RESERVES.

# Every rental funds its next repair.



A workshop ledger that separates rental income, refundable deposits, and the money that keeps each item working.

MAKERSPACES · CAMPUS HUBS · TOOL LIBRARIES

# The tool is shared.
# The repair budget usually isn’t.

Repeated use creates wear. The workshop needs a clear answer to three practical questions.

01

### Who pays for the next repair?

Rental revenue can be spent before a tool breaks. Maintenance needs its own protected balance.

02

### What happened to the deposit?

Damage claims need recorded evidence, clear approval rules, and a visible refund deadline.

03

### Can members check the record?

Operators need a history that connects payments, returns, decisions, and repairs to an item.

ONE ITEM. ONE CONTINUING REPAIR RESERVE.

# Scan. Rent. Return.
# Keep the equipment in use.

01 / REGISTER

## Give each tool a QR

Set its fee, deposit, reserve share, and distinct arbiter.

02 / RENT

## Fund the next repair

The fee splits into operator income and this item’s reserve.

03 / RETURN

## Record the evidence

A photo commitment starts the dispute window and Cronpot job.

04 / REFUND

## Close the deposit

The keeper executes the due job; contract rules enforce eligibility.

05 / RESOLVE

## Review a damage claim

The operator’s proposal needs renter or arbiter approval.

06 / REPAIR

## Pay from the reserve

An arbiter-approved repair spends the item’s own repair funds.

THREE BALANCES. DIFFERENT PERMISSIONS.

# A deposit is not income.
# A repair reserve stays with its item.

OPERATOR INCOME

## 80%

Of the rental fee in the demo configuration. Withdrawable by the operator.

ITEM REPAIR RESERVE

## 20%

Of the rental fee in the demo configuration. Spent only on approved repairs.

REFUNDABLE DEPOSIT

## Separate

Held as a rental liability. Returned in full or reduced by an approved claim.

Illustrative split, not a completed live payment. Human assessment still determines damage; unresolved votes can delay settlement.

RUNTIME PRIMITIVES, WITH MEASURED BEHAVIOR

# Solidity rules.
# Native accounts and evidence.

PALLET-REVIVE + SUBSTRATE SIGNING

## Use community wallets

Substrate-signed Revive calls run Solidity accounting and permissions. V3 funding automatically maps native accounts to H160.

14 NATIVE / 18 EVM DECIMALS

## Keep value exact

The funded P4 probe measured a 10,000 conversion factor. Integer conversions reject fractional native-unit dust.

TRANSACTIONSTORAGE

## Commit the final photo bytes

A signed 1 KiB store succeeded in P6. SHA-256 checks the displayed bytes; a persistent backup outlives native retention.

CRONPOT + FINALIZED NATIVE EVENTS

## Reusable timed execution

MeterMend integrates a generic timed-call registry that other contracts can reuse. SQLite retains finalized history beyond node pruning.

KEEPER MODE SELECTED ON V3

# Automatic execution. Contract-enforced refund authority.

“refunds execute automatically via Cronpot; execution is permissionless and our keeper runs it.”

01

## Return

MeterMend records the deadline and schedules a Cronpot job.

02

## Wait

The configured keeper observes finalized chain time.

03

## Execute

Any eligible caller can execute the due Cronpot job.

04

## Refund

MeterMend checks status and deadline before releasing a deposit.

**What the funded probe found:** the native Scheduler rejected our signed origin with `BadOrigin` (P5, block 213576). It is not the refund executor in this deployment. Cronpot supplies liveness; the contract retains authority.

METERMEND.VERCEL.APP/PROOF ↗

# Both contracts finalized. The live rental proof comes next.

| | | |
|---|---|---|
| **Ping contract**<br>Finalized · 213570 | `0x96ee88d272550530b518cbaa15a80cf5d38cdcfd`<br>Runtime deployment and signed value movement verified | `tx 0x5003af97…24a011c` ↗ |
| **P4 / P6 probes**<br>Finalized | Exact 14/18 value movement; signed 1 KiB native store<br>Blocks 213573 and 213579 · Full receipts retained | Funded evidence ↗ |
| **Cronpot**<br>Finalized · 265131 | `0x41635a685448cfa86da1600656f103a211e398f4`<br>Original deployment transaction finalized after the node resumed. | `tx 0xd60b3e13…1a0df93` ↗ |
| **MeterMend**<br>Finalized · 271379 | `0x61382db168db646b899234cd707c7ba0142c4071`<br>Both contract read-backs passed. Live rental, refund, dispute and repair proof pending. | `tx 0x23b8aa6f…62b8f17` ↗ |

OPEN SOURCE · MIT ↗

# Built to preserve the record when a connection fails.

**43**

## Contract tests

Accounting, reserve isolation, dispute votes, callback failures, and fuzzed fee splits.

Unchanged Foundry suite; previously passed in CI.

**137**

## TypeScript tests

Chain handling, durable services, transaction recovery, web flows, and deployment scripts.

October 31 baseline; package and script checks passed.

**29**

## Browser checks

Desktop/mobile layout, wallet states, photo integrity, and recovery after reload.

Synthetic recovery fixtures; no payment was submitted.

Durable transaction hashes | Account locks across tabs | Offline retained history | Production build passes

TESTS SUPPORT THE IMPLEMENTATION; THEY DO NOT REPLACE THE PENDING LIVE GOLDEN PATH

PROPOSED MODEL · PILOT VALIDATION AHEAD

# Earn when shared equipment is used and maintained.

PROPOSED REVENUE

## A small service fee per completed rental.

Price for the workshop’s operating budget. Keep refundable deposits and protected repair money separate.

The current testnet contract does not collect a MeterMend platform fee. Pricing and collection are future product work.

### Makerspaces

Shared tools, membership communities, and repair accountability.

### Campus hubs

Student equipment, shared media kits, and transparent handovers.

### Tool libraries

Repeated lending with a visible item-by-item maintenance history.

# Make the next repair part of every rental.

NOW / SUBMISSION

## Finish the live proof

Run the golden path, activate persistent services, deliver Telegram updates, and record the actual demo.

NEXT / PILOT

## Learn in workshops

Test onboarding, dispute handling, reserve adequacy, evidence retention, and willingness to pay with real operators.

THEN / ECOSYSTEM

## Reuse Cronpot

Document timed-call integration, improve keeper operations, and pursue DeBox as the target community integration.

**Support needed:** reliable testnet access, a persistent operations host, and pilot workshop partners.

metermend.vercel.app ↗

github.com/willy264/portal_dot ↗